

A working Verlinde formula for logarithmic CFT

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Outline

- 1 The Verlinde formula in rational CFT
- 2 Log CFT vs rational CFT
- 3 Working Verlinde formula in log CFT by example

The basics

- Correlators, most important quantity in CFT:

$$\langle \psi_1(z_1) \cdots \psi_n(z_n) \rangle$$

- Fusion product: Given two reps $R_1 \ni \psi_1, R_2 \ni \psi_2$, what is the largest rep $R_3 \ni \psi_3$ such that

$$\langle \psi_1(z_1) \psi_2(z_2) \psi_3(z_3) \rangle$$

is non-zero?

The basics

- Fusion products very tedious to compute honestly (NGK algorithm)
- For rational CFT: Fusion given by Verlinde formula

The Verlinde formula

[Zhu]

For rational CFT, characters of reps

$$\text{ch}[R](\tau) = \text{Tr}_R(q^{L_0 - c/24}), \quad q = e^{2\pi i \tau}$$

admit action of $\text{SL}(2, \mathbb{Z})$

$$\begin{aligned} \text{ch}[R_i](\tau + 1) &= \sum_j T_{i,j} \text{ch}[R_j](\tau) \\ \text{ch}[R_i](-1/\tau) &= \sum_j S_{i,j} \text{ch}[R_j](\tau). \end{aligned}$$

Here R_j irred.

The Verlinde formula

[Zhu], [Moore, Seiberg], [Huang]

Denote fusion product by

$$R_i \times R_j = \sum_k N_{i,j}^k R_k.$$

The Verlinde formula is

$$N_{i,j}^k = \sum_l \frac{S_{i,l} S_{j,l} \overline{S_{k,l}}}{S_{0,l}}.$$

Here “0” is index of vacuum rep.

Example: Yang-Lee model at $c = \frac{-22}{5}$

- Two irreps

l_0 conf. weight 0, $l_{-1/5}$ conf. weight $\frac{-1}{5}$

- S-matrix

$$S = \begin{pmatrix} -\frac{\sqrt{2}}{\sqrt{5-\sqrt{5}}} & \frac{\sqrt{3-\sqrt{5}}}{\sqrt{5-\sqrt{5}}} \\ \frac{\sqrt{3-\sqrt{5}}}{\sqrt{5-\sqrt{5}}} & \frac{\sqrt{2}}{\sqrt{5-\sqrt{5}}} \end{pmatrix}$$

- Fusion rules

$$l_0 \times l_i = l_i, \quad i = 0, \frac{-1}{5}$$
$$l_{-1/5} \times l_{-1/5} = l_0 \oplus l_{-1/5}$$

Log CFT vs rational CFT

[Gurarie], [Kausch], [Flohr]

- Virasoro zero mode can have Jordan blocks
- Correlators have log singularities

Problems:

- Characters can't distinguish all reps

$$0 \rightarrow A \longrightarrow B \longrightarrow C \rightarrow 0$$

$$\text{ch}[B](\tau) = \text{ch}[A](\tau) + \text{ch}[C](\tau)$$

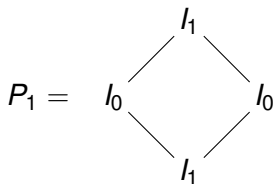
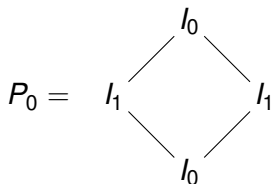
- S-matrix becomes τ -dependent

Resolve these issues by example.

$c = -2$ simplest example of a log CFT

[Gaberdiel, Kausch]

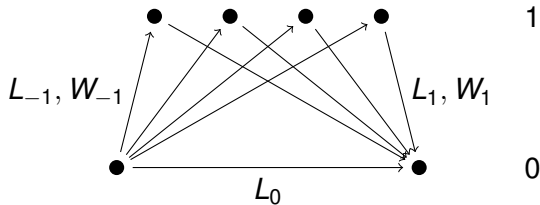
- Generated by T and 3 W -fields of weight 3
- Irreds: $l_0, l_1, l_{-1/8}, l_{3/8}$
- Indecomposables:



- Characters: $\text{ch}[P_0](\tau) = \text{ch}[P_1](\tau) = 2\text{ch}[l_0](\tau) + 2\text{ch}[l_1](\tau)$

$c = -2$ simplest example of a log CFT

The indecomposable P_0 :



$$L_0 = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

$c = -2$ simplest example of a log CFT

[Gaberdiel, Kausch]

Fusion rules:

$\times$	I_0	I_1	$I_{-1/8}$	$I_{3/8}$	P_0	P_1
I_0	I_0	I_1	$I_{-1/8}$	$I_{3/8}$	P_0	P_1
I_1	I_1	I_0	$I_{3/8}$	$I_{-1/8}$	P_1	P_0
$I_{-1/8}$	$I_{-1/8}$	$I_{3/8}$	P_0	P_1	$2I_{-1/8} \oplus I_{3/8}$	$2I_{-1/8} \oplus I_{3/8}$
$I_{3/8}$	$I_{3/8}$	$I_{-1/8}$	P_1	P_0	$2I_{-1/8} \oplus I_{3/8}$	$2I_{-1/8} \oplus I_{3/8}$
P_0	P_0	P_1	$2I_{-1/8} \oplus I_{3/8}$	$2I_{-1/8} \oplus I_{3/8}$	$2P_0 \oplus 2P_1$	$2P_0 \oplus 2P_1$
P_1	P_1	P_0	$2I_{-1/8} \oplus I_{3/8}$	$2I_{-1/8} \oplus I_{3/8}$	$2P_0 \oplus 2P_1$	$2P_0 \oplus 2P_1$

- Fusion does not close on irreds
- Fusion closes on above list of reps

$c = -2$ simplest example of a log CFT

[Gaberdiel, Kausch]

S -matrix is τ -dependent

$$\text{ch}[l_0]\left(\frac{-1}{\tau}\right) = \frac{1}{4}\text{ch}[l_{-1/8}](\tau) - \frac{1}{4}\text{ch}[l_{3/8}](\tau) - \frac{i\tau}{2}\text{ch}[l_0](\tau) + \frac{i\tau}{2}\text{ch}[l_1](\tau)$$

Potential resolutions:

- Enlarge set of characters to recover genuine $\text{SL}(2, \mathbb{Z})$ rep
- Restrict/cut down chiral algebra. Study transformations of characters of reps of smaller algebra.

Working Verlinde formula in log CFT by example

[Creutzig, Ridout], [Adamović, Milas]

Singlet algebra at $c = -2$. Generated by T and 1 W -field of weight 3

- Obtained by infinite order orbifold
- There exist irreps I_h^s for any real conformal weight $h \geq \frac{-1}{8}$
- For $h > \frac{-1}{8}$ there always exist two inequivalent irreps
- Convenient parametrisation of reps by free field reps F_λ ,
 $h_\lambda = \frac{1}{2}\lambda(\lambda - 1)$, $\lambda \in \mathbb{R}$

More on free field reps

- F_λ reducible singlet rep iff $\lambda \in \mathbb{Z}$
- Irreds $I_{h_\lambda}^s$ with conf. weight $h_\lambda = \frac{1}{2}\lambda(\lambda - 1)$, $\lambda \in \mathbb{Z}$ are precisely those that can appear in indecomposables.
- Characters of irreds with $h_\lambda = \frac{1}{2}\lambda(\lambda - 1)$, $\lambda \in \mathbb{Z}$ are infinite sums of free field characters, e.g. $\lambda = 0$

$$I_0^s = \sum_{n \geq 0} F_{-n} - F_{-n-\frac{1}{2}}$$

Transformation of free field characters

Characters of free field reps F_λ transform very simply

$$\text{ch}[F_\lambda]\left(\frac{-1}{\tau}\right) = \int_{\mathbb{R}} S[F_\lambda \rightarrow F_\rho] \text{ch}[F_\rho](\tau) d\rho$$

$$S[F_\lambda \rightarrow F_\rho] = e^{-2\pi i(\lambda - \frac{1}{2})(\rho - \frac{1}{2})}$$

Continuous version of Verlinde formula

[Creutzig, Ridout]

Reps R_1, R_2

$$N_{R_1, R_2}^{F_\nu} = \int_{\mathbb{R}} \frac{S[R_1 \rightarrow F_\rho] S[R_2 \rightarrow F_\rho] \overline{S[F_\nu \rightarrow F_\rho]}}{S[I_0^s \rightarrow F_\rho]} d\rho$$

If $I^s[\lambda]$, $\lambda \in \frac{1}{2}\mathbb{Z}$ is the irred subrep of F_λ , then the Verlinde fusion rules are

$$\begin{aligned} I^s[n] \times I^s[m + \frac{i}{2}] &= I^s[n + m + \frac{i}{2}] \\ I^s[n + \frac{1}{2}] \times I^s[m + \frac{1}{2}] &= I^s[m + n] + I^s[m + n + 1] \end{aligned}$$

$n, m \in \mathbb{Z}, i = 0, 1.$

- By a process called simple current extension, the triplet can be reassembled from the singlet algebra.
- The induced fusion rules for the triplet algebra match the real fusion rules on the level of characters!

$\times$	l_0	l_1	$l_{-1/8}$	$l_{3/8}$
l_0	l_0	l_1	$l_{-1/8}$	$l_{3/8}$
l_1	l_1	l_0	$l_{3/8}$	$l_{-1/8}$
$l_{-1/8}$	$l_{-1/8}$	$l_{3/8}$	$2l_0 \oplus 2l_1$	$2l_0 \oplus 2l_1$
$l_{3/8}$	$l_{3/8}$	$l_{-1/8}$	$2l_0 \oplus 2l_1$	$2l_0 \oplus 2l_1$

Conclusion

- In log CFT the S -matrix becomes τ -dependent
- The Verlinde formula can be saved by considering characters of a subalgebra (infinite order orbifold)
- The example presented is the simplest example of a large family of theories. This family includes the $c = 0$ case.

The End

Thank you for your attention!